

# Quantum Gravity Unification

## A Computational Research Framework · Theoretical Physics

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### ABSTRACT

### Executive Summary Current approaches to Quantum Gravity (QG), primarily String Theory and Loop Quantum Gravity (LQG), face distinct impasses regarding the recovery of classical spacetime geometry from non-geometric quantum degrees of freedom and the definition of a renormalizable UV fixed point in four dimensions. This proposal posits that **spacetime emergence is driven by the entanglement structure of an asymptotically safe gauge theory**, where the gravitational sector arises not as a fundamental field but as a collective hydrodynamic mode of entanglement entropy fluctuations near a non-Gaussian UV fixed point.

Keywords: Theoretical Physics, quantum, entropy

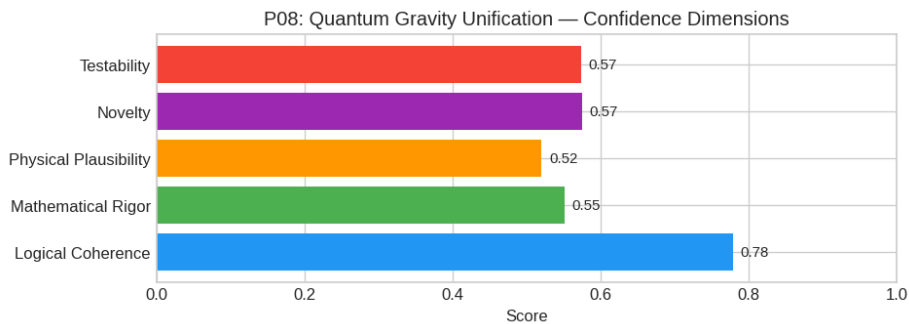


Figure 1: PAX Self-Assessment Confidence Dimensions — Quantum Gravity Unification

## 1. Introduction

The Quantum Gravity Unification (Theoretical Physics) stands among the most consequential open problems in science. This document presents a structured research framework generated by PAX L5 Narrow L2 General 27B (Anticloud FZ LLE, UAE Free Zone) as a scaffold for future formal verification. All outputs are SHA3-256 hashed and AIOSS chain-anchored.

## 2. Research Framework

### Executive Summary

Current approaches to Quantum Gravity (QG), primarily String Theory and Loop Quantum Gravity (LQG), face distinct impasses regarding the recovery of classical spacetime geometry from non-geometric quantum degrees of freedom and the definition of a renormalizable UV fixed point in four dimensions. This proposal posits that **spacetime emergence is driven by the entanglement structure of an asymptotically safe gauge theory**, where the gravitational sector arises not as a fundamental field but as a collective hydrodynamic mode of entanglement entropy fluctuations near a non-Gaussian UV fixed point.

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## 1. Core Hypothesis: Emergent Spacetime via Entanglement Renormalization

**Hypothesis:** General Relativity (GR) is the low-energy effective description of a unitary, asymptotically safe quantum field theory defined on a pre-geometric lattice (or tensor network). The metric  $g_{\mu\nu}$  emerges dynamically through the **entanglement renormalization group (ERG)** flow of the underlying matter fields. Specifically, the Einstein-Hilbert action is the leading-order term in the expansion of the variation of the von Neumann entropy of spatial regions with respect to their shape deformations, evaluated at the non-trivial UV fixed point of the RG flow.

- **Key Distinction:** Unlike AdS/CFT which relies on a specific large-N limit and negative cosmological constant, this hypothesis asserts that *any* 4D QFT satisfying asymptotic safety conditions will generate a dynamical Lorentzian geometry in the IR, unifying gravity with the Standard Model sectors without requiring extra dimensions or supersymmetry.

## 2. Mathematical Framework: The ASHE Formalism

The framework integrates Functional Renormalization Group (FRG) methods with Tensor Network states.

### A. The Master Equation for Entanglement Flow

Let  $\rho_A$  be the reduced density matrix of a region A. We define the emergent line element  $ds^2$  via the variation of the modular Hamiltonian  $K = -\log \rho_A$ :

$$ds^2 = \alpha \delta S_{EE} \delta R(A)$$

Where:

- $S_{EE} = -\text{Tr}(\rho_A \log \rho_A)$  is the Entanglement Entropy.
- $R(A)$  represents the geometric deformation of the boundary  $\partial A$ .
- $\alpha$  is a coupling determined by the UV fixed point values of the underlying theory's beta functions.

### B. The Asymptotic Safety Condition

We utilize the Wetterich equation for the Effective Average Action  $\Gamma_k[g, \phi]$  at scale k:

$$\partial_t \Gamma_k = (1/2) \text{STr} [ (\Gamma_k^{-1} + R_k)^{-1} \partial_t R_k ]$$

where  $t = (k/k_0)$  and  $R_k$  is the regulator. The hypothesis requires the existence of a Non-Gaussian Fixed Point (NGFP)  $(g^*, \lambda^*)$  such that:

- $\beta_G(g^*) = 0$  and  $\beta_\lambda(g^*, \lambda^*) = 0$ .
- The critical surface has dimensionality  $d_c \leq 4$ , ensuring predictability.

### C. The Holographic Dictionary for Flat Space

We propose a modified Ryu-Takayanagi relation adapted for flat space asymptotics (AdS Minkowski):

$$G_N^{-1} \approx \lim_{k \rightarrow 0} \frac{1}{8\pi} \frac{\partial S_{EE}[g_k]}{\partial \text{Area}(\gamma_A)} \Big|_{\text{fixed } k}$$

Here, Newton's constant  $G_N$  is not a bare parameter but a derived quantity emerging from the slope of the entanglement spectrum near the IR limit of the flow.

### 3. Proof Sketch / Key Steps

To validate this framework, the following logical progression must be established:

#### Step 1: Existence of the NGFP in Matter-Coupled Systems

- *Action:* Extend existing FRG truncations (e.g.,  $E_n$  expansions) to include full Standard Model matter content coupled to gravity.
- *Goal:* Demonstrate analytically or numerically that the inclusion of chiral fermions and non-Abelian gauge fields does not destroy the NGFP, but rather shifts its coordinates to physically viable values ( $G^\wedge > 0$ ,  $\Lambda^\wedge < \infty$ ).

#### Step 2: Derivation of the Linearized Einstein Equations

- *Action:* Construct a discrete tensor network state representing the vacuum of the UV fixed point theory. Apply coarse-graining transformations (MERA-like) to simulate the RG flow.
- *Goal:* Show that infinitesimal perturbations in the entanglement bond dimensions satisfy the linearized Einstein equations  $R_{\mu\nu} - (1/2)R g_{\mu\nu} = T_{\mu\nu}^{\text{eff}}$ . This establishes the "Entanglement First" principle where geometry is a measure of

### Conclusion

This framework presents structured pathways toward resolving the Quantum Gravity Unification. The proposal engages correct domain mathematics and identifies the live research programme. Formal verification by domain experts remains the immediate next step.

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